Harnoor Kaur Sagar

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EDUCATION

McMaster University Sept. 2024 - June 2029
B.Eng., Software Engineering (Co-op) *GPA: 3.74/4.0*
Awards: Dean's Honour List (2024-25), DECA Alumni Scholarship (**$15,000**), Award of Excellence, Top 10% (**$3,000**)
Relevant Courses: Data Structures and Algorithms, Object-Oriented Programming, Software Development, Discrete Math

TECHNICAL SKILLS

Languages: Python, Java, TypeScript, JavaScript, SQL, C, C#, HTML, CSS
Frameworks and APIs: Flask, React, Node.js, Express, JUnit, Google Apps Script, REST APIs, LLM APIs, RAG
Developer Tools: Git, GitHub Actions, GitLab, Jira, Google BigQuery, Apache Airflow, SonarQube, Netlify, Linux, Figma

EXPERIENCE

Geotab May 2026 - Present
Product Safety Analytics Intern, Embedded Engineering *Oakville, ON*

- Engineered an **LLM**-powered hazard-analysis app in **Python**, **Flask**, and **JavaScript** with **parallel requirement extraction**, **severity scoring**, and **embedding-based deduplication** across **5** product categories.
- Built a **RAG** pipeline cross-referencing product docs against **ISO/SAE** standards and internal safety policy, detecting **80%+** of gaps with **90%+** citation accuracy and cutting report time **50%+**.
- Fixed **silent reporting failures** in an internal analytics backend by tracing the **call chain** and implementing **deterministic hashing** and **chunked caching**.
- Built a **full-stack** internal **Apps Script** portal with **account-scoped access control**, scaling fleet outreach to **thousands** of customer databases and replacing **months** of manual coordination.
- Co-built an **agentic Airflow** incident-response tool that drafts **GitLab** fixes, cutting data analysts' failed-pipeline investigation from **120 to 30 min**.
- Debugged vehicle interoperability via **CAN bus** fingerprinting and **debug log/payload** analysis across vehicle protocols, tracing a **fleet-wide** device anomaly to its **root cause**.

Google Developer Groups McMaster June 2025 - Present
Community and Code Team Lead *Hamilton, ON*

- Led a **10+ member** team to host **10+** technical workshops and professional development events for **500+** students.
- Engaged **industry professionals** as panelists for **resume roasts** and **system design** challenges with co-leads.

McMaster Formula Electric Feb. 2025 - Present
Website Developer *Hamilton, ON*

- Redesigned the website in **React**, **HTML**, and **CSS**, reducing page load times by **80%** and improving responsiveness.
- Implemented **CI/CD** deployment pipelines through **Netlify**, enabling automated builds and faster, reliable updates.

McMaster Engineering Venture Academy May 2025 - Aug. 2025
Engineering Instructor Intern *Hamilton, ON*

- Built and debugged **embedded systems** with **sensors**, **LEDs**, and **motor controls** for **150+** students in grades 3-12.
- Performed **cost-benefit analysis** of project materials, reducing expenses by ∼**15%** without lowering quality.

PROJECTS

Kashio (ElleHacks 2026) | Code | Devpost *TypeScript, React, Node.js, Express, ElevenLabs API*

- Built a **full-stack** financial literacy game in **TypeScript**, **React**, and **Node.js** for the **Wealthsimple** category.
- Integrated **ElevenLabs text-to-speech** with **Google Translate** via an **Express** endpoint, narrating in **6** languages.
- Designed a **learn-by-doing** game loop where kids run virtual businesses, take **loans with interest**, and earn boosts by answering money questions in a racing mini-game.

Catan Simulator | Code *Java, JUnit, GitHub Actions, SonarQube*

- Developed a **Java** Catan simulator with a rule-enforcing **game engine**, **3** AI agents, and **JSON** state export.
- Applied **object-oriented design patterns** to decouple game rules, player actions, and rendering, enabling **undo/redo** and **extensible** rule enforcement.
- Automated quality gates with a **GitHub Actions CI** pipeline running builds and **SonarQube** static analysis on every push and pull request, backed by **31 JUnit** unit tests.